

BMEN 600 Midterm Research Plan

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Introduction and Motivation

Determining whether an increase in healthcare expenditure is positively correlated with an increase in life expectancy across various countries, social factors, and variations in gender, age, and ethnicity is a question many healthcare systems have been attempting to answer. Most, if not all, governments and their associated healthcare systems allocate limited resources to meet their population's health needs. The association of whether expenditure and life expectancy positively correlate with one another has been observed to vary depending on factors such as how healthcare is financed (public vs. private), individual/population income, education, and other social and health factors [1]. Furthermore, it has previously been observed that higher healthcare spending does not guarantee better care and extended life expectancy [2]. Understanding the differences in what factors can lead to better healthcare outcomes, measured by increased life expectancy, is crucial to making decisions about health-care funding, particularly in countries with limited resources and unmet health needs.

Our team seeks to investigate whether higher health-care expenditure per capita is positively associated with various categories of life expectancy across countries outlined in the World Health Organization (WHO) Global Health Observatory Data and Global Health Expenditure Data spanning 21 years (from 2000 to 2021). Expenditure will be measured in Purchasing Power Parity (PPP) to account for differences in currency valuation between countries and equalize the purchasing power of each country [3]. Life expectancy at birth is determined by the number of years a newborn would be expected to live based on the observed mortality rates in that year [4].

An expected key uncertainty in this study is how the association between healthcare expenditure and life expectancy is influenced by variations in private and public spending, as well as other social factors. To address this, additional analyses will be performed with outcomes from WHO's World Health Statistics indicators, including but not limited to, neonatal and under-five mortality rates per 1000 live births and maternal mortality ratio per 100,000 live births [5]. By examining life expectancy alongside various social factors, this study will assess whether higher health-care spending is consistently associated with better population health. Understanding where these associations differ can help inform discussions about health-care funding and the limits of using expenditure alone to explain differences in health outcomes.

Background and Research Gap

Current research investigating the relationship between healthcare expenditure and life expectancy utilizes national statistics, international comparisons, and models that follow countries over time. Ray and Linden analyzed 195 countries from 1995 to 2014 using dynamic panel models and expenditure data with sources from the WHO [1]. Their study showed that public expenditure showed more favorable associations with health outcomes compared to private expenditure; however, results were limited to the specific model used in the study. Using quantile regression across 175 countries from 2006 to 2011, Obrizan and Wehby found that healthcare expenditure

was more strongly associated with life expectancy in countries where life expectancy was lower, as such their findings suggest that the strength of the relationship varies across countries [6].

Studies focusing on high-income countries also show that higher health-care spending does not always correlate to longer life expectancy. Using data mainly from 2013 to 2016, Papnicolas et al. found that the United States had higher healthcare spendings but lower life expectancy than other high-income countries [2]. The authors identified higher salaries for health-care workers, pharmaceutical prices, and administrative costs as major contributors to higher US spending. These findings highlight that the amount spent as well as the costs of providing care should be considered when comparing countries and outcomes. Roffia et al. examined 36 Organisation for Economic Co-operation and Development (OECD) countries from 1999 to 2018. Their regression analysis and dynamic models demonstrated that higher health expenditure was associated with longer life expectancy at lower spending levels, but the association became negative beyond an estimated turning point [7]. Moreover, their findings linked life expectancy with not only expenditure, but also several health systems and social factors. However, because the study focused only on OECD countries, its findings may not be applicable to lower-income countries with different health-care needs and resources.

Other studies have considered how spending is allocated and financed by the government. Bradley et al. analysed pooled data from 30 OECD countries between 1995 and 2005 and found that countries with higher social spending relative to health-care spending were observed to have longer life expectancy. This association remained after accounting for differences in GDP per capita and healthcare expenditure [8]. In a study conducted by Gabani et al. to examine differences between health financing systems, a transition towards government financing was associated with improvements in life expectancy relative to predominantly private financing [9].

Our study will utilize WHO data from 2000 to 2021 to examine the association between healthcare spending and life expectancy across countries. We will further categorize our results using data published in the World Health Statistics 2026 from the WHO based on health conditions and social factors. By comparing these outcomes, we will assess whether higher spending is consistently associated with longer life expectancy, or whether the strength of the association differs on other factors. These findings will describe patterns observed in individual countries and regions, providing outcomes that can be used to guide policy and shape decisions in healthcare systems.

Dataset and Responsible Use

This research project uses publicly available, country-level data from the WHO accessed on October 9th, 2026. Life expectancy at birth was obtained from the WHO's Global Health Estimates, and current health expenditure per capita was obtained from the Global Health Expenditure Database (GHED). Life expectancy is expressed in years, while expenditure is expressed in current purchasing power parity (PPP) international dollars per person. The GHED compiles national health accounts, government expenditure records, official statistics, and estimates where necessary. The estimation methods are documented in *WHO methods and data*

sources for life tables 2000-2021 [10] and *Methodology for the update of the Global Health Expenditure Database* [11]. We will be conducting further research on life expectancy based on WHO regions and variable death factors using the dataset documented in *World health statistics 2026: monitoring health for the SDGs, sustainable development goal* [5].

The downloaded expenditure dataset includes 195 countries and territories. After combining the two datasets by country and year, and removing regional and worldwide totals, the final dataset includes 183 countries and territories from 2000-2021. Each row represents one country-year observation. Each country or territory constitutes a distinct analytical unit, with repeated annual measurements that cannot be treated as independent observations. The principal variables are country, year, total-population life expectancy at birth, and current health expenditure per capita. For the analysis of changes over time, the independent variable will be calendar year (2000–2021), and the dependent variable will be each country's ratio of life expectancy at birth to health expenditure per capita. This ratio will be expressed as years of life expectancy per 1,000 PPP international dollars spent per person on health care. For each country, the independent variable will be the year, and the dependent variable life expectancy divided by health expenditure expressed as years per 1,000 PPP international dollars per person.

The use and sharing of the dataset will follow the applicable WHO terms, including attribution, preservation of source information, and avoidance of misrepresentation or implied WHO endorsement. The WHO life-expectancy data, Global Health Expenditure Database, and World Health Statistics data are open access sources through WHO's public data platforms and can be downloaded without registration or a formal data-access application. This will allow the team to obtain the data directly for the proposed analysis. All data sources will be appropriately cited, and access dates and applicable data-use terms will be documented. Potential limitations include missing observations, differences in national reporting quality, and reliance on modeled mortality estimates where registration systems are incomplete. Also, excluding countries with unavailable data may introduce selection bias, and current-price expenditure is not an inflation-adjusted spending measure. The results obtained in this study will describe relationships between countries, without claiming that higher spending causes individual longer life expectancy or that the observed relationship applies to all individuals.

Proposed Methods and Evaluation

The analysis uses a country-year dataset containing five key variables. These variables are country, year, life expectancy at birth for the total population, current health expenditure per capita (PPP adjusted), and a derived efficiency metric representing life expectancy per 1000 PPP dollars of healthcare spending. These measurements allow comparisons across countries and over time, supporting both longitudinal trend analysis and cross-sectional evaluation of healthcare system efficiency. The data has been cleaned by first validating the presence of required columns within the time range between the years 2000 and 2021. This includes removing duplicate country-year entries and converting all numeric fields into appropriate types. Once the presence of all data is

confirmed, the next pre-processing step recalculates the efficiency ratio using the formula $1000\ X\ \frac{life\ expectancy}{expendeture}$. If the expenditure value is not valid, the ratio is left blank to avoid the use of an incorrect data point. Planned analytical steps include descriptive statistics, visualization of temporal trends, cross-country and cross-region comparisons, correlation analysis between expenditure and life expectancy, and regression modelling to quantify relationships or efficiency differences. All work will be conducted in Python within Google Colab, using Pandas for data manipulation, NumPy for numerical operations, and additional libraries such as Matplotlib, Seaborn, Statsmodels, or Scikit-learn if visualization or modelling is performed.

Currently, the code that has been developed by the team extracts and plots health expenditure vs. life expectancy as well as life expectancy per health expenditure over time for each WHO country. This code was developed to look for overall data trends to guide the future development of the future code. Future development of the code will start region specific data comparisons to make the data comparisons more manageable. Within the specific regions, code will need to be developed to run statistical analysis or different variables available within the WHO dataset selected. Statistical comparisons will be run on relevant variables to look for trends within and between countries and to guide future investigations. The data produced will then lead to statistics cells which will run relevant statistical tests based on variables that will be chosen in future meetings. Statistical tests that may be used include Pearson and Spearman correlations, regression models or group comparison tests such as t-tests or ANOVAs. Finally, this data will be fed into statistical plotting cells to produce final visual outputs.

The main result we will use to answer our question of interest will be based on the direction and strength of the correlation between healthcare expenditure and life expectancy. Based on similar studies, we expect there to be a positive correlation found between healthcare expenditure and life expectancy, where variables such as Gross Domestic Product (GDP), number of doctors, and air pollution can have a positive effect on life expectancy [12].

Our results will appropriately address the research question as the direction and strength of this correlation could indicate whether greater healthcare expenditure is associated with longer life expectancy, when several factors previously mentioned are also factored in. Our results will be compared to previous studies such as Anwar et al. (2023) where they examined the impact of health expenditures on health outcomes in the OECD countries [12]. To place our results in context, we will compare our findings with Zarulli et al. (2021), who used output-oriented data envelopment analysis to assess life expectancy achieved at given healthcare expenditure levels [13]. While our correlation analysis will examine whether higher expenditure is associated with longer life expectancy, their approach provides context for understanding why countries with similar expenditure may achieve different health outcomes.

To avoid producing misleading results, our initial plan is to treat each country as a single unit with repeated yearly observations. We will calculate correlations separately for each year to assess whether the association between healthcare expenditure and life expectancy is consistent over time. This will prevent accounting for repeated measurements within the same countries in the analysis.

To show variability, scatter plots will be used at country-level observations with a 95% confidence interval and the number of countries included. If missing values are present, they will be excluded from the analysis. Due to the influence of variables such as income, education, and healthcare access, it will be clarified that correlation does not establish causation.

Preliminary Results

The matched dataset includes 183 countries and territories; most countries have complete data for all 21 years, but missing expenditure values indicate that the number of countries included varies over the selected years. It was seen that a variation exists in expenditure and life expectancy across countries, with each point representing one country-year observation, and repeated observations from the same country (Appendix, Figure 1). A general positive association between healthcare expenditure and life expectancy is depicted, with countries with higher spending showing longer life expectancy (Appendix, Figure 1). However, once individuals reach the 70-year mark, it appears that countries with similar expenditure show varying life expectancies (Appendix, Figure 1). This preliminary result reveals an interesting point of investigation for our dataset, which is the potential influence of income, education, and other social factors along with healthcare access, that may be acting to facilitate this variation in life expectancy observed (Appendix, Figure 1). To address this preliminary result, our future steps will include determining how the factors listed may be impacting the trends observed, as well as accounting for such factors in our research question.

We have also analyzed the ratio of countries' life expectancy over health expenditure over a 21-year period, to determine if they varied notably across the period in question (Appendix, Figure 2). Overall, findings indicate that the majority of countries have been experiencing a steady decline in the ratio of life expectancy over health expenditure in the past 2 decades, but potential reasons as to why this may be occurring, or if this decline is significant, remains to be investigated in our study (Appendix, Figure 2). The potential conclusions that can be drawn from our dataset and future findings can help answer and narrow down the question of how countries can most efficiently allocate their expenditure to optimize healthcare outcomes indicated by increased life expectancy. We will work towards answering this research question by analyzing if the factors previously listed will potentially interfere with life expectancy (income, education, social determinants), from countries or regions deemed to be displaying optimal results.

As part of our final investigation, the number of countries included in each reported year must be determined, and it must be assessed whether missing or excluded data affects the findings. We will use statistical models and comparisons to analyze the relationship(s) between our factors and variables of interest and determine how to best present comparisons to build a cohesive story to our findings. Our findings will also need to be compared with previous studies to ensure the reproducibility of our work.

Teamwork and Project Plan

Table 1. Project plan detailing allocation of the major tasks needed to complete the project with members assigned as leads or collaborators for each task.

Task	Lead(s)	Collaborators and Reviewers	What does the whole team need to understand?
Literature Review and Background Research	Tala, Irene	Rest of the team	Key findings, knowledge gaps, and relevance to our research question.
Dataset Selection/Evaluation	Jessica, Irene	Rest of the team	Meaning of variables, data source, population, and limitations.
Dataset Cleaning/Preprocessing	Brendan, Irene	Rest of the team	What was changed, why, and potential effects on results.
Analysis (statistics, machine learning, etc.)	Brendan, Jessica	Rest of the team	Methods used, rationale, assumptions, and limitations.
Validation and Evaluation of Findings	Irene, Tala	Rest of the team	How reliability and performance will be assessed.
Interpretation and Discussion	Tala, Jessica	Rest of the team	What results mean, limitations, and connection to the research question.
Documentation and Reproducibility	Jessica, Brendan	Rest of the team	Where methods, code, data decisions, and results are documented.

Timeline of major tasks to be completed:

- Literature Review and Background Research: October 9th – October 16th
- Dataset Selection/Evaluation: October 2nd – October 16th
- Dataset Cleaning/Preprocessing: October 2nd – Ongoing
- Analysis (statistics, machine learning, etc.): October 9th - Ongoing
- Validation and Evaluation of Findings: October 9th - Ongoing
- Interpretation and Discussion: October 9th - Ongoing
- Documentation and Reproducibility: October 2nd – Ongoing
- Presentation: November 15th – Due November 30th or December 4th
- Final Write up Preparation and Completion: Ongoing – Due December 4th

References

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Appendix

Figures

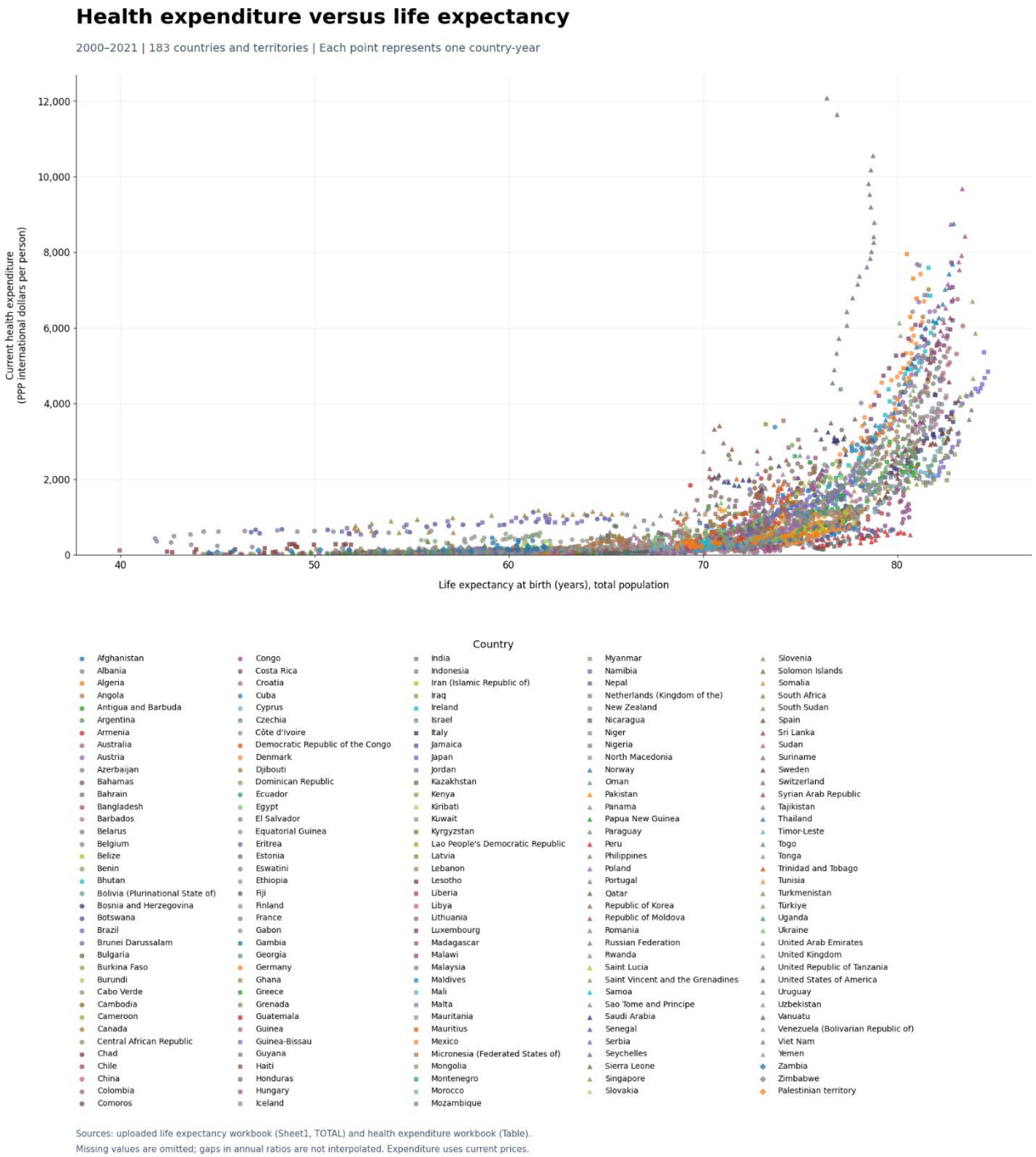


Figure 1. Life expectancy at birth (years) of the total reported population versus current health expenditure (PPP international dollars per person) for each WHO country (183 countries and territories) from 2000-2021.

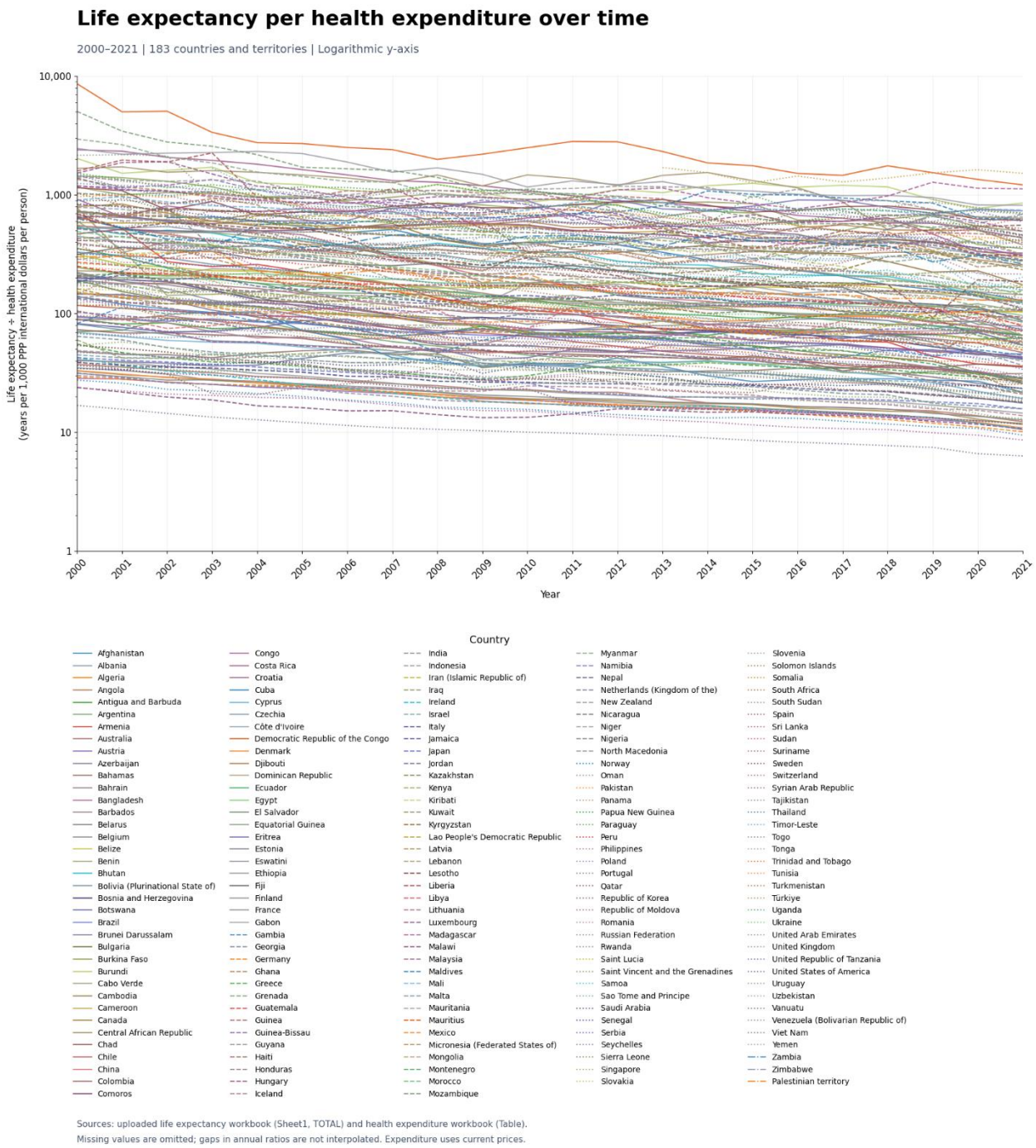


Figure 2. The ratio of life expectancy over health expenditure (years per 1000 PPP international dollars per person) of the total reported population for each WHO country (183 countries and territories) per year from 2000-2021.

GitHub Repository Link: https://github.com/blnfisher/BMEN-600-Team-12-Term-Project-Repository-/tree/main